Linear Algebra and Analytic Geometry

Unit 1: Matrices and Systems

Worked Solutions to Exercises 1–34

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November 3, 2026

1. **Indicate which of the following matrices are in echelon form and which are in reduced echelon form.**

a) $A = \begin{bmatrix} 1 & 1 & 5 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Echelon form, not reduced. The pivots move strictly to the right, but the second pivot is -1, not 1.

b) $B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 5 & 1 & 1 \end{bmatrix}$

Neither. The first nonzero entries of the rows are in columns 1, 2 and 1, respectively.

c) $C = \begin{bmatrix} 1 & 5 & -4 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The second and third rows have their first nonzero entry in the same column.

d) $D = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Echelon form, not reduced. The pivots are in positions $(1,1)$ and $(2,3)$, but $d_{13} = -4$ is nonzero above the second pivot.

e) $E = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are 1, move strictly to the right and have only zeros above them. The zero row is at the bottom.

f) $F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are in positions $(1,1)$ and $(2,3)$, both equal 1, with zeros above them. The zero row is at the bottom.

g) $G = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The first row is zero, but there are nonzero rows below it.

h) $H = \begin{bmatrix} 0 & 2 & 4 & 0 \\ 0 & 0 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Echelon form, not reduced. The pivots are in positions $(1, 2)$ and $(2, 3)$, and the zero rows are at the bottom. However, the pivots are 2 and 8, not 1.

i) $I = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Neither. The second row is zero, but the third is not.

j) $J = \begin{bmatrix} 1 & 5 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. There is only one pivot, equal to 1, and all remaining rows are zero.

k) $K = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The pivots are in positions $(1, 1)$ and $(2, 4)$, both equal 1, with zeros above them. The zero rows are at the bottom.

l) $L = \begin{bmatrix} 0 & 1 & -7 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. The only pivot is $l_{12} = 1$, and all remaining rows are zero.

m) $M = \begin{bmatrix} 0 & 1 & -7 \end{bmatrix}$

Reduced echelon form. The only pivot is $m_{12} = 1$. The remaining conditions hold trivially.

n) $N = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Neither. The first two rows have their first nonzero entry in the same column.

o) $O = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Reduced echelon form. All rows are zero. Since there are no pivots, the conditions concerning them hold vacuously.

2. Final item. Remove when done.